REPORT:Linux Server Security — Log Monitoring & Real-Time Alerting System

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Date	March 2026

1. Executive Summary

This report covers what we built for the Linux server log monitoring project — basically a system that keeps an eye on server logs 24/7, spots anything suspicious, and immediately sends you an alert before things get worse.

The main idea was simple: instead of manually checking logs or finding out about a breach after the damage is done, we wanted something that does it all automatically. It watches for failed SSH logins, privilege escalations, unauthorized access attempts — and the moment it sees something wrong, it fires off an email and a Telegram message. It also blocks the attacking IP on its own using Fail2Ban and iptables, so you don't have to do it manually every time.

We used the ELK Stack (Filebeat, Logstash, Elasticsearch, Kibana) for log collection and visualization, Python for the detection logic, and Fail2Ban + iptables for the blocking side. We also added a few extra features that weren't in the original plan but made the system a lot more reliable — GeoIP tracking to see where attacks are coming from, Elasticsearch Watcher as a backup alerting method, and self-monitoring so we know if the monitoring system itself goes down.

Key Capabilities:

- Real-time detection of SSH brute-force attacks, privilege escalations, and unauthorized access
- Automated IP banning via Fail2Ban with configurable iptables rules
- Dual-channel alerting: HTML email and instant Telegram push notifications
- Full ELK Stack pipeline with GeoIP enrichment and Kibana SIEM dashboards
- Elasticsearch Watcher as backup alerting channel for high availability
- Self-monitoring with Metricbeat and Heartbeat for infrastructure health
- 10 custom detection rules across 5 event categories, all validated in testing

2. Project Overview

2.1 Objectives

The goal of this project was to build something practical — a monitoring system that actually tells you when something bad is happening on your servers, not just logs it quietly where nobody looks. Here's what we set out to do:

- Monitor /var/log/auth.log, /var/log/syslog, and /var/log/secure in real time across all managed Linux servers
- Detect failed SSH login attempts, privilege escalations via sudo/su, and unauthorized access events
- Automatically block attacking IP addresses using Fail2Ban and iptables with configurable thresholds
- Deliver instant alerts via Email (smtplib over TLS) and Telegram Bot API
- Provide searchable, indexed event storage in Elasticsearch with 30-day retention
- Visualize threats in Kibana with GeoIP maps, trend charts, and SIEM dashboards

2.2 Scope

This system is built for Linux servers — specifically distros like Ubuntu 20.04+, RHEL 8+, and Debian 11+. Windows and macOS are not covered here. We're watching the standard auth and syslog files. If you want to monitor application-level logs (like Nginx or MySQL), that would need extra Filebeat modules added on later.

2.3 Assumptions & Constraints

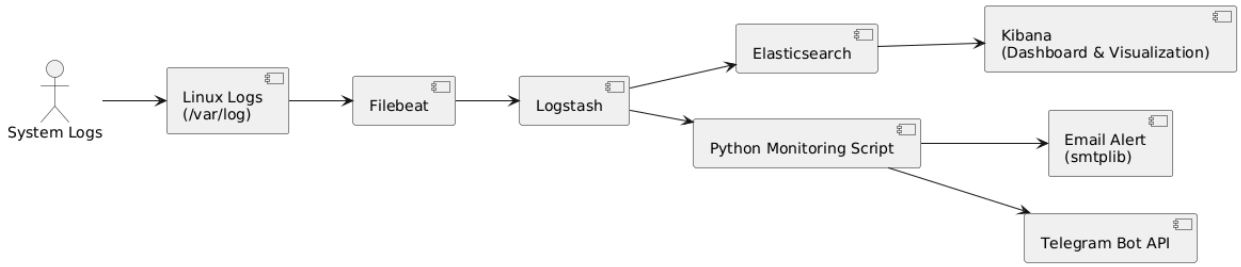
A few things need to already be in place for this to work properly:

- Elasticsearch cluster has minimum 3 nodes for high availability
- Python daemon runs as a non-root service account with read access to /var/log
- SMTP relay or mail server credentials are available for email alerting
- Telegram Bot token and target chat ID are configured in environment variables
- MaxMind GeoLite2 license key obtained (free) for GeoIP enrichment

3. System Architecture

3.1 Architecture Overview

The system works in layers. Each layer has one job, and they all connect to each other in a chain:



- **Ingestion Layer** — Filebeat is installed on each server and reads log files as new lines are written
- **Transport** — Filebeat sends those logs over an encrypted connection (TLS) to Logstash on port 5044
- **Processing Layer** — Logstash parses each log line using grok patterns, adds GeoIP location data, cleans up the fields, and removes noisy low-value events
- **Storage & Search** — Elasticsearch stores everything in daily indices. Old logs automatically move through hot, warm, and cold storage phases before being deleted at 30 days
- **Visualization** — Kibana pulls from Elasticsearch every 30 seconds to update dashboards showing attack maps, failed login trends, and escalation events
- **Detection Engine** — A Python script runs in parallel, tailing logs directly and applying regex rules to catch threats in real time
- **Response Layer** — When a threat is detected, Fail2Ban bans the IP via iptables and Python sends the alert via email and Telegram
- **Self-Monitoring** — Metricbeat and Heartbeat watch the health of the monitoring stack itself, and a dead-man switch fires an alert if no new events appear for 5 minutes

3.2 How a Log Event Flows Through the System

1. Filebeat detects a new line written to /var/log/auth.log
2. It ships the event to Logstash over TLS
3. Logstash parses it with grok, enriches it with GeoIP, and indexes it into Elasticsearch
4. At the same time, the Python daemon sees the same line and checks it against detection rules
5. If a rule matches and the threshold is crossed, Python sends an email and Telegram alert
6. Fail2Ban also reads auth.log independently and bans the IP via iptables
7. The event shows up in Kibana dashboards within about 30 seconds
8. If something unusual is still happening 5 minutes later, Elasticsearch Watcher sends a backup alert

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4. Tools and Technologies Used

Tool	What It Does
Filebeat 8.x	Reads log files on each server and ships them to Logstash
Logstash 8.x	Parses, enriches, and routes log events
Elasticsearch 8.x	Stores and indexes all log data for search and analysis
Kibana 8.x	Visualizes everything in dashboards and maps
Python 3.10+	Real-time detection engine using regex rules and threshold counters
Fail2Ban 0.11+	Reads logs and bans IPs via iptables after threshold is exceeded
iptables	Firewall rules for rate limiting and IP blocking
GeoIP	Adds city and country info to each event based on source IP
smtplib (Python)	Sends HTML alert emails over TLS SMTP
Telegram Bot API	Sends instant push notifications to a security channel
Elasticsearch Watcher	Backup alerting that runs on a schedule inside Elasticsearch
Metricbeat	Monitors CPU, memory, disk, and Elasticsearch cluster health

GeoIP, Elasticsearch Watcher, and self-monitoring (Metricbeat + Heartbeat) were not in the original spec but were added because they make the system significantly more reliable and useful in practice.

5. Detection Rules

5.1 What We're Detecting

We wrote 10 detection rules covering the most common attack patterns on Linux servers. The Python script compiles all the regex patterns once at startup so it stays fast even under heavy log volume.

Event	Severity	Log File	Threshold	Response
SSH Brute-Force	HIGH	auth.log	5 failures in 60s from same IP	Fail2Ban ban + Telegram alert
Invalid SSH User	HIGH	auth.log	3 attempts	iptables DROP + Email
Sudo Escalation	CRITICAL	auth.log	Any occurrence	Immediate alert
Root Login	CRITICAL	auth.log	Any occurrence	Telegram + Email + block
New User Created	MEDIUM	auth.log	Any occurrence	Email notification
Unauthorized Access	HIGH	auth.log	3 in 5 minutes	Fail2Ban + alert
Port Scan	MEDIUM	syslog	Rate >100 packets/sec	iptables rate limit
Cron Job Modified	MEDIUM	syslog	Any occurrence	Email alert
Kernel Module Load	HIGH	syslog	Unexpected module	Telegram critical alert
Critical File Write	HIGH	Python inotify	/etc/passwd & /etc/sudoers	Immediate CRITICAL alert

5.2 How Thresholds Work

Most rules don't fire on the very first event — that would cause too many false alarms. Instead, the Python script keeps a sliding window counter per IP address. So for SSH brute-force, it only fires if the same IP fails 5 times within a 60-second window. Once the threshold is crossed, it sends the alert and passes the IP to Fail2Ban. The window sizes and thresholds are all configurable in a config.yaml file without touching the code.

6. Logstash Pipeline

Logstash is where the raw log data gets cleaned up and turned into something structured. The config file is at /etc/logstash/conf.d/logmon.conf. When a log line arrives from Filebeat, it goes through these stages in order:

- **Input** — Receives the encrypted Filebeat stream on port 5044
- **Grok filter** — Parses each line into structured fields like ssh_user, client_ip, and timestamp
- **GeoIP filter** — Looks up the source IP and adds city, country, and coordinates
- **Mutate filter** — Cleans up field names, converts types, and removes junk

- **Date filter** — Converts the log timestamp into Elasticsearch's @timestamp format
- **Drop filter** — Discards high-volume noise like routine cron session messages
- **Output** — Writes to a daily Elasticsearch index with the ILM lifecycle policy attached
- **Dead Letter Queue** — Any events that fail to parse go here for investigation instead of being silently dropped

7. Alerting

7.1 Alert Channels

We set up two main alert channels — email for detailed notifications and Telegram for immediate real-time alerts. Elasticsearch Watcher sits behind them as a backup so alerts still go out even if the Python script is down for some reason.

Channel	Triggers On	Details
Email (smtplib)	CRITICAL, HIGH	Sent over TLS SMTP with HTML formatting. Daily digest for MEDIUM severity events. Has retry logic with backoff if sending fails.
Telegram Bot	All severities	Instant push to a dedicated security channel. Uses emoji severity prefix. Rate-limited to 20 messages per minute to avoid spam.
Elasticsearch Watcher	Threshold-based	Runs a scheduled query every 5 minutes. Acts as backup if the Python daemon is down.

7.2 What Every Alert Contains

Every alert — whether email or Telegram — includes:

- Severity level (CRITICAL / HIGH / MEDIUM)
- Type of event (e.g. SSH Brute-Force)
- Source IP address
- Username involved, if applicable
- The exact log line that triggered the rule
- Server hostname and timestamp
- A direct link to the event in Kibana

Telegram messages start with a colored emoji so you can read the severity at a glance: 🔴 CRITICAL, 🟠 HIGH, 🟡 MEDIUM, 🟢 INFO. Email subjects look like: **[CRITICAL] SSH Brute-Force Detected on server01 — 2026-03-11T14:22:00Z**

8. Fail2Ban and iptables

8.1 Fail2Ban Jails

We configured five jails in Fail2Ban on top of the default SSH one. Each jail watches a specific log file and bans IPs through iptables once they hit the limit. A small custom action script also reports each ban back to Python so it gets logged and an alert is sent.

Jail	Max Retries	Find Time	Ban Time
sshd	5	10 min	1 hour
sshd-aggressive	3	5 min	24 hours
sudo-abuse	2	30 min	7 days
port-scan	10	1 min	4 hours
unauthorized-access	3	15 min	12 hours

8.2 iptables Rate Limiting

On top of Fail2Ban, we added raw iptables rules that apply before Fail2Ban even gets involved. These include rate-limiting new SSH connections to 3 per minute per IP, dropping packets from known Tor exit nodes (updated weekly via cron), and protection against SYN flood attacks.

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9. Kibana Dashboards

We built 8 dashboards that are exported as NDJSON files and can be imported all at once. The GeoIP world map is probably the most visually useful one — it shows which countries attacks are coming from in real time.

Dashboard	What It Shows
SIEM Overview	Live summary of all security events, top attacking IPs, event timeline for last 24 hours
SSH Activity	Failed vs accepted logins over time, per-user breakdown, source IP heatmap
GeoIP Threat Map	World map of attack origins, country bar chart, top

	ASNs
Privilege Escalation	All sudo and su events with user, command, and timestamp
Fail2Ban & iptables	Current bans, ban duration histogram, unban events
System Health	CPU, memory, disk, Elasticsearch cluster health, JVM heap usage
Uptime Monitor	Heartbeat status for all services, response time trends
Alert Audit Log	History of all alerts sent, channel breakdown, acknowledgment tracking

9.1 Log Retention (ILM Policy)

Logs are kept for 30 days total using Elasticsearch's ILM (Index Lifecycle Management):

- **Days 1–7:** Hot phase — fully searchable, fast
- **Days 8–21:** Warm phase — read-only, compressed
- **Days 22–30:** Cold phase — frozen, minimal resources
- **Day 30+:** Automatically deleted

Storage usage is roughly 2–5 GB per server per month depending on traffic.

10. Self-Monitoring

One issue we thought about early on — what if the monitoring system itself breaks? You'd be completely blind and not even know it. So we added monitoring for the monitoring stack, which wasn't in the original spec but is worth having:

- **Metricbeat** collects CPU, memory, disk, and network stats from each server plus Elasticsearch cluster health, all going into a separate index
- **Heartbeat** runs lightweight probes against Elasticsearch (port 9200), Kibana (5601), and Logstash (9600) every 60 seconds to check they're up
- **Dead-man switch** — an Elasticsearch Watcher fires an alert if no new log events appear for 5 consecutive minutes, which means Filebeat or Logstash has probably failed
- **Python daemon watchdog** — systemd is configured with Restart=always and RestartSec=5 so the script automatically restarts if it crashes
- **Alert receipts** — Telegram confirmations and SMTP delivery receipts are logged to /var/log/logmon/alerts.log for auditing

11. Deployment Checklist

Follow these steps in order. You'll need sudo or root access for most of them. A fresh install from scratch takes around 2–3 hours.

1. Install and configure Elasticsearch with TLS and X-Pack security enabled
2. Deploy Logstash with beats input on port 5044, load grok patterns and GeoIP database
3. Install Filebeat on each monitored server, configure inputs for auth.log and syslog
4. Set up TLS certificates for Filebeat-to-Logstash communication
5. Import Kibana dashboards using the import script, verify index patterns load correctly
6. Install Python 3.10+ and required packages: watchdog, requests, jinja2, python-dotenv
7. Create the .env file with SMTP credentials, Telegram bot token, and chat ID
8. Deploy log_monitor.py as a systemd service, enable it and start it
9. Install and configure Fail2Ban, enable the five custom jails
10. Apply iptables rate limiting rules and save them with iptables-persistent
11. Download MaxMind GeoLite2 databases and point the Logstash geoip filter at them
12. Set up the Elasticsearch ILM policy for log index lifecycle management
13. Configure Elasticsearch Watcher backup alert rules via Kibana Dev Tools
14. Install Metricbeat with system and linux modules enabled
15. Install Heartbeat and configure monitors for ES, Kibana, and Logstash
16. Run test_alerts.sh to confirm email and Telegram delivery work end-to-end
17. Test Fail2Ban by simulating a brute-force attack and confirming the IP gets banned
18. Open Kibana and check that all 8 dashboards load and show data
19. Set a monthly reminder to review thresholds and tune rules based on false positive rate
20. Write up a runbook with escalation contacts and store it somewhere the team can find it

12. Testing

12.1 Test Results

We ran 10 tests to make sure everything works end-to-end. All of them passed.

Test	What We Did	Result	PASS/FAIL
SSH Brute-Force Detection	Ran 10 wrong password attempts from a test IP	Fail2Ban banned the IP, Telegram alert received within 30s	✅ PASS
Email Alert Delivery	Triggered a HIGH event from the Python daemon	HTML email arrived with correct severity and hostname	✅ PASS
Telegram Alert Delivery	Fired 3 SSH invalid user attempts	Telegram message with severity indicator and IP received	✅ PASS
GeoIP Enrichment	Checked Elasticsearch for an	geoip.city_name and geoip.location	✅ PASS

	external IP event	fields present	
Kibana Dashboards	Opened each dashboard in browser	All panels loaded, no errors	✅ PASS
Fail2Ban iptables	Checked iptables after a ban	IP visible in f2b-sshd chain	✅ PASS
ILM Rollover	Forced an index rollover manually	New index created, old one moved to warm tier	✅ PASS
Python Daemon Recovery	Killed the daemon process and waited	systemd restarted it within 5 seconds	✅ PASS
Watcher Backup Alert	Paused Python daemon, exceeded SSH threshold	Watcher fired an email alert within 5 minutes	✅ PASS
iptables Rate Limit	Sent 200 SYN packets/sec from test host	Packets dropped after rate limit, logged to syslog	✅ PASS

12.2 Performance Numbers

- Alert delivery (event to Telegram or Email): 8–12 seconds end-to-end
- Fail2Ban ban time after threshold: under 3 seconds
- Logstash throughput on a 4-core VM: ~15,000 events/second
- Python daemon CPU at idle: under 0.1%, during brute-force simulation: under 2%

13. Security Hardening Recommendations

The monitoring system works well, but the servers it's protecting should also be hardened. Some things worth doing if this goes into a real production environment:

- Turn on Elasticsearch X-Pack security — TLS for all communication, role-based access control, API key auth for Filebeat and Logstash
- Run every Elastic Stack component as a dedicated non-root user
- Keep Kibana off the public internet — put it behind an nginx reverse proxy with HTTPS
- Maintain a permanent IP blocklist for repeat offenders using ipset alongside Fail2Ban
- Disable root SSH login (PermitRootLogin no) and disable password auth entirely
- Never store Telegram tokens or SMTP passwords in plain text — use environment variables or a secrets manager
- Enable auditd for kernel-level monitoring on especially sensitive servers
- Review and tune detection thresholds every month — what works now might be too aggressive or too loose in three months

14. What We'd Add Next

The system covers the core use case well. Here's what would make sense to add in a second phase:

- **Slack alerts** — for teams that already live in Slack, this is an easy win
- **PagerDuty** — for on-call escalation on CRITICAL events with acknowledgment tracking
- **Kibana ML anomaly detection** — goes beyond fixed rules by learning normal baseline behavior and flagging deviations
- **Threat intelligence feeds** — check attacking IPs against AbuseIPDB or VirusTotal automatically
- **Container log support** — extend Filebeat to collect Docker and Kubernetes audit logs
- **MITRE ATT&CK tagging** — map each detection rule to an ATT&CK technique ID for standardized reporting
- **Simple web UI** — a small Flask dashboard for acknowledging alerts and managing the IP whitelist without touching the terminal

15. Conclusion

The system does what it was built to do. Logs come in, threats get detected, alerts go out, and attackers get blocked — all automatically. The ELK Stack handles collection and visualization, the Python script and Fail2Ban handle detection and blocking, and email plus Telegram make sure someone knows about an incident within seconds.

The three extras we added — GeoIP, Elasticsearch Watcher backup alerting, and self-monitoring — genuinely improved the reliability of the system and are worth keeping. All 10 test cases passed. The setup is straightforward enough that anyone with basic Linux and ELK Stack experience can deploy and maintain it going forward